

Dhaivat Naik

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PROFESSIONAL EXPERIENCE

Analyst 2 – Information Systems, *The State of New Jersey*

Jul '25 – Present

- Built **FastAPI** microservices, containerized with **Docker** and deployed on **Kubernetes (EKS)**, to replace manual file transfers, reducing data availability delays from 3+ hours to real-time
- Prototyped a **Kafka + MongoDB** event pipeline to automate inter-departmental data sharing, to reduce bottlenecks from batch processing of **10K+** departmental updates daily

Research Assistant, *Rutgers Institute for Health, Healthcare Policy and Aging Research*

Nov '23 – Oct '24

- Developed a scalable ETL pipeline using **Python** to extract **750,000** health survey records, transformed datasets using **NumPy** and **Pandas** processing **256** columns and stored the data into **BigQuery** using **GCP's** native Python APIs
- Conducted statistical analysis on extracted data for policy recommendation using **SPSS**, **R**, and **MATLAB** by identifying patterns through regression, clustering, and hypothesis testing, showing how patient responses are linked to specific health conditions
- Implemented **MERF** using **scikit-learn** to analyze healthcare data, incorporating both fixed and random effects to reduce prediction error by **20%** (RMSE) compared to a standard random forest model

Data Engineering Intern, *SuitUp*

Jun '24 – Aug '24

- Extracted sales data using **Salesforce SOQL** into AWS **Glue**, automating schema mapping into **Glue Data Catalog**, reducing schema validation time from 5 to ~2.5 hrs
- Performed analysis using **PySpark** on **20M** records from Amazon **S3** to identify key sales metrics such as customer churn rate and purchase intervals, leading to a **12%** improvement in customer retention through targeted ads and personalized offers
- Streamlined the migration of **200 GB** transactional data from on-premise Oracle to **Redshift** using AWS **DMS** and Lambda triggers, reducing duration by **50%**

Data Scientist I, formerly Data Science Intern, *SAS Institute*

Jan '22 – Jul '23

- Contributed to the deployment of **12** prototypes for BFSI clients, using predictive modeling and data visualization, driving **\$1.2M+** in revenue
- Developed customer demos using **SAS Viya** for machine learning model deployment and **Kubernetes** for containerized management, leveraging **Microsoft Azure** for scalable infrastructure and cloud storage
- Engineered data pipelines using **SAS Data Integration Studio** and **Databricks**, integrating **50K+** daily transactional records into analytics systems
- Collaborated with Customer Advisory team to design and present interactive dashboards using **SAS Visual Analytics** and **PowerBI** to clients, visualizing customer segments, geo-analytics, and sales performance, securing a **\$300,000** RFP

Software Development Intern, *PMaps*

Jun '21 – Dec '21

- Collaborated with senior developers to optimize performance of a distributed **Apache Airflow** pipeline, improving partitioning strategies on **Kafka** and reducing data retrieval time by ~2s on the user dashboard
- Deployed unit and integration test cases using **PyTest**, and **Docker** for containerization, increasing code coverage and addressing critical edge cases, resulting in a **25%** reduction in reported production bugs
- Developed reusable UI components using **Angular**, with a **RESTful API** layer to connect to **AWS RDS** to dynamically fetch and display data, which streamlined end-to-end interactions

PROJECTS

LoanShield – Real Time Loan Risk Scoring Platform

Mar '25 – May '25

- Built a scalable machine learning pipeline on **GCP** to predict loan defaults using LendingClub data (2M+ accepted, 27M+ rejected loans)
- Engineered datasets in **BigQuery** by integrating raw loan data from Cloud Storage buckets and cleaning ~150 features, applying partitioning by issue date and clustering by loan status to reduce data scanned per query using **Python** in **Vertex AI Workbench**
- Executed supervised learning using **Logistic Regression**, **Random Forest**, and **XGBoost**, achieving an **AUC-ROC** of **0.87** and **F1** score of **0.74**

FineD-AI-ne Restaurant QA System

Sept '24 – Dec '24

- Developed a restaurant **Q&A** system using **LLMs (Mistral, Llama3, T5)** and **RAG architecture** that processes natural language customer queries across **10+** key categories (menu, pricing, ambiance, dietary accommodations, wait time, service, etc)
- Combined Yelp and Google reviews with semantic matching and sentiment analysis to generate contextual responses, processing **170+** questions per restaurant and achieving **84%** semantic accuracy in providing restaurant information

TweetX Tracker

Mar '24 – May '24

- Developed a real-time Twitter sentiment analysis app using **Python**, **Flask**, **Databricks**, and the **Twitter API** across **2M+** tweets
- Analyzed sentiment with **NLP** techniques and **TextBlob**, and used **Plotly** for dynamic visualizations of tweet trends and sentiment insights

SKILLS

- **Languages:** Python, SQL, Scala, R, Java, Base SAS, C++, CASL, Swift
- **Database Technologies:** PostgreSQL, MySQL, Snowflake, NoSQL, MongoDB, Cassandra, DynamoDB
- **Cloud Platforms:** AWS, GCP, Azure, Salesforce, SAS Viya
- **Tools & Libraries:** Apache Spark, Kafka, Kubernetes, Delta Lake, PySpark, Hive, Hadoop, TensorFlow, MLlib, Keras, PyTorch, Koalas, Azure DevOps, GitLab, Docker, Spring Boot, Angular, REST API, Tableau, PowerBI, PyTest, Postman, Jira, MLOps
- **Data Engineering:** ETL, Streaming, Warehousing, Integration, Modeling, Dimensional Modeling, SCD, CDC, CI/CD
- **Coursework:** Natural Language Processing, Big Data Testing, Causal Inference, DSA, Predictive Modeling, Visualization, Distributed System

EDUCATION

Rutgers University

Master of Science in Data Science – Statistics

Narsee Monjee Institute of Management Studies

Bachelor of Technology in Computer Engineering